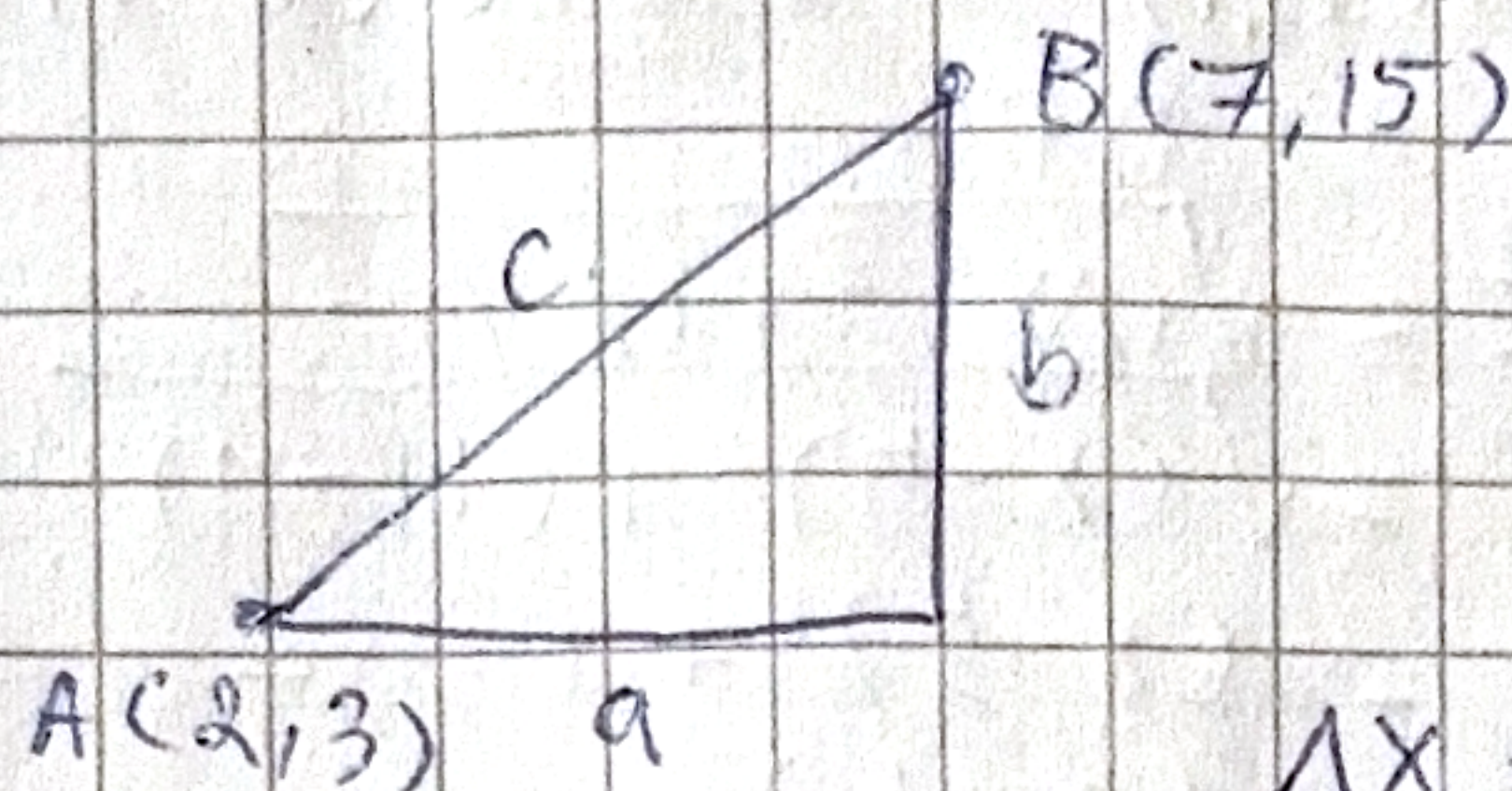


SECTION A: DISTANCE & MIDPOINT

1. FIND THE DISTANCE BETWEEN POINTS
A(2,3) AND B(7,15)



$$\begin{aligned}\text{DISTANCE } c^2 &= a^2 + b^2 \\ &= (\Delta x)^2 + (\Delta y)^2 \\ &= \sqrt{(\Delta x)^2 + (\Delta y)^2}\end{aligned}$$

$$\begin{aligned}\Delta x &= x_2 - x_1 = x_B - x_A \\ &= 7 - 2 = 5\end{aligned}$$

$$\begin{aligned}\Delta y &= y_2 - y_1 = y_B - y_A \\ &= 15 - 3 = 12\end{aligned}$$

$$\begin{aligned}\text{DISTANCE (d)} &= \sqrt{(5)^2 + (12)^2} \\ &= \sqrt{25 + 144} = \sqrt{169} \\ &= 13\end{aligned}$$

THE DISTANCE BETWEEN A AND B IS 13

2. FIND THE MIDPOINT OF THE SEGMENT JOINING
C(-4,6) AND D(8,-2)

$$\begin{aligned}\text{MIDPOINT} &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\ &= \left(\frac{(-4) + 8}{2}, \frac{6 + (-2)}{2} \right) \\ &= \left(\frac{4}{2}, \frac{4}{2} \right) = (2, 2)\end{aligned}$$

THE MIDPOINT OF C AND D IS (2,2)

3. A POINT $P(x, 4)$ IS EQUIDISTANCE FROM $A(1, 1)$ AND $B(5, 1)$, FIND x

$$\text{DISTANCE } P(x, 4) \rightarrow A(1, 1) = \text{DISTANCE } P(x, 4) \rightarrow B(5, 1)$$

$$DPA = DPB$$

$$\text{DISTANCE} = \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

$$DPA \quad \sqrt{(x-1)^2 + (4-1)^2} = \sqrt{(x-5)^2 + (4-1)^2} \quad DPB$$

$$(x-1)^2 + (3)^2 = (x-5)^2 + (3)^2$$

$$P \rightarrow A = P \rightarrow B \quad (x-1)^2 + 9 = (x-5)^2 + 9$$

$$a^2 + 2ab + b^2 = \dots$$

$$(x-1)^2 = (x-5)^2$$

$$x-1 = x-5$$

$$x$$

$$(x^2 + 2(x)(-1) + 1) + ((4)^2 + 2(4)(-1) + 1)$$

$$\sqrt{(1-x)^2 + (1-4)^2} = \sqrt{(5-x)^2 + (1-4)^2}$$

$$x^2 + (-2x) + 1 + 16 + (-8) + 1$$

$$x^2 - 2x + 10$$

$$= x^2 + 2x(-5) + 25 + 16 + 2(-4) + 1$$

$$= x^2 - 10x + 25 + 16 - 8 + 1$$

$$x^2 - 2x + 10$$

$$= x^2 - 10x + 34$$

$$-2x + 10x$$

$$= 34 - 10$$

$$8x$$

$$= 24$$

$$x$$

$$= 3$$

SECTION B : SLOPE & LINES

4. FIND THE SLOPE OF THE LINE THROUGH $(3, -2)$ AND $(-1, 6)$

$$\begin{aligned}\text{SLOPE} &= \Delta y / \Delta x \\ &= (y_2 - y_1) / (x_2 - x_1) \\ &= (6 - (-2)) / ((-1) - 3) \\ &= 8 / (-4) = -2\end{aligned}$$

SLOPE IS -2 .

5. WRITE THE EQUATION OF THE LINE PASSING THROUGH $(2, 5)$ WITH SLOPE -3

EQUATION OF LINE $y = mx + b$.

$y = 5, x = 2, m(\text{SLOPE}) = -3$

INSERT y, x, m $5 = (-3)(2) + b$

$$b = 5 - (-6)$$

$$= 11$$

THE EQUATION IS $y = (-3)x + 11$.

6. FIND THE EQUATION OF THE LINE PASSING THROUGH $(1, 2)$ AND $(4, 11)$ IN SLOPE-INTERCEPT FORM.

EQUATION OF LINE $y = mx + b$.

FIND m (SLOPE)

$$\text{SLOPE} = \Delta y / \Delta x$$

$$= (11 - 2) / (4 - 1) = 9/3$$

$$= 3$$

FIND b . ($y = mx + b$)

POINT $(1, 2)$ $2 = (3)(1) + b$

$$b = 2 - 3$$

$$= -1$$

POINT $(4, 11)$ $11 = 3(4) + b$

$$b = 11 - 12$$

$$= -1$$

EQUATION OF LINE IS $y = 3x - 1$.